Ammar Kheder

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PROFILE

Creator of Otso (ammarkheder.com/Otso), a real-time global air-pollution forecast app at 1 km resolution, updated daily. Doctoral researcher in computational engineering at LUT University (AMC-Lahti, INAR), combining atmospheric physics with generative networks. Lead author of Foehn, the model behind Otso (preprint submitted to Nature, 2026), and of TopoFlow (npj Climate and Atmospheric Science, 2026). Large-scale distributed training on the LUMI supercomputer (up to 1,024 AMD MI250X GPUs). Freelance AI research expert with Mercor.

EDUCATION

2024 - now — **Ph.D., Computational Engineering**
LUT University, Lahti, Finland
Atmospheric Modelling Centre (AMC-Lahti). Supervisors: Prof. Michael Boy, Assoc. Prof. Zhi-Song Liu.

2021 - 2024 — **M.Sc., Engineering (Big Data and Artificial Intelligence)**
EiCnam Paris, France
Apprenticeship at INRIA Bordeaux (Mnemosyne team).

2019 - 2021 — **B.Sc., Data Science (BUT STID)**
BUT Niort, Universite de Poitiers, France
Statistics, big data, business intelligence, data science.

EXPERIENCE

2026 - now — **AI Research Expert (freelance)**
Mercor, San Francisco (remote)
Scientific data-analysis workflows (Python, ML, atmospheric and climate data) used to train AI agents that operate computers. Top contributor tier within two weeks.

2024 - now — **Junior Researcher and Teaching Assistant**
LUT University, Computational Engineering
Research on generative, physics-aware deep learning for atmospheric science. Teaching assistant for Foundations of AI and Machine Learning (practical sessions and exam grading, twice) and Foundations of Computer Science; co-supervision of one master's thesis a year. Contributor to the AMC-Lahti summer school (25 participants, 2025).

2023 - 2026 — **CEO and Founder, Wabel Group**
AI services and web development
Built cDESP, a B2B SaaS compliance platform for industrial refrigeration pressure equipment, in production for DCRR.

2021 - 2022 — **Research Engineer (Apprenticeship)**
INRIA Bordeaux, Mnemosyne team
Worked on LingoRob, a crowdsourcing platform for multi-language linguistic corpora used to train Reservoir Computing models for human-robot interaction. Supervisor: Xavier Hinaut.

2021 - 2023 — **Volunteer Firefighter**
Sapeurs-Pompiers des Deux-Sevres (SDIS 79), France
2 years 8 months of on-call service. Certifications: FISPV, PSC 1, PSE 1, PSE 2, SAP, INC 1, DIV.

PUBLICATIONS

Global probabilistic forecasting of air pollution (Foehn): 1 km PM2.5 forecasts up to a week ahead over all inhabited land.
A. Kheder, M. Boy, Z.-S. Liu.
Preprint, 2026. arXiv:2610.25941. Interactive forecasts, film and code at ammarkheder.com/Foehn

TopoFlow: topography-aware pollutant Flow learning for high-resolution air quality prediction.
A. Kheder, H. Toropainen, W. Peng, S. Antao, J. Chen, M. Boy, Z.-S. Liu.
npj Climate and Atmospheric Science 9, 143 (Nature Portfolio), 2026. DOI: 10.1038/s41612-026-01417-5

CRAN-PM: Cross-Resolution Attention Network for High-Resolution PM2.5 Prediction.
A. Kheder et al.
Preprint, 2026 (under review). arXiv:2603.11725

Climate downscaling with likelihood functional flow matching.
Z.-S. Liu, H. Toropainen, A. Kheder, M. Boy.
Preprint, 2026. arXiv:2610.33482

Nowcasting aerosol particle size distributions from PM2.5 with physics-regularized flow matching.
Z.-S. Liu, J. Koykka, W. Peng, A. Kheder, P. Clusius, M. Boy.
Preprint, 2026. arXiv:2610.33557

An actively trained neural operator for near-explicit atmospheric chemistry.
Z.-S. Liu, W. Peng, A. Kheder, R. Makkonen, M. Boy.
Preprint, 2026. arXiv:2610.33771

Inverse Neural Operator for ODE Parameter Optimization.
Z.-S. Liu, W. Peng, H. Toropainen, A. Kheder, A. Rupp, H. Froning, X. Lin, M. Boy.
Preprint, 2026. arXiv:2603.11854

Deep Spatio-Temporal Neural Network for Air Quality Reanalysis.
A. Kheder, B. Foreback, L. Wang, Z.-S. Liu, M. Boy.
Scandinavian Conference on Image Analysis (SCIA 2025), Springer LNCS. DOI: 10.1007/978-3-031-95911-0_6

TALKS AND PRESENTATIONS

- IAC 2026 (International Aerosol Conference), Xi'an, China (September 2026): TopoFlow.
- Boya Forum on Environmental Research, Peking University (August 2026): invited talk.
- EAC 2025 (European Aerosol Conference), Lecce, Italy. AQ-Net poster.
- SCIA 2025, Reykjavik, Iceland. Speed talk and poster, Springer LNCS publication.
- LUT Doctoral School Science Conference 2025 (20 May), Lappeenranta.
- Invited visit to BUT SD Niort (5 January 2026): academic and professional journey.

SELECTED PROJECTS

- Otso (2026): real-time global air-pollution forecast at 1 km, updated daily on LUMI, available in all languages as a web app and PWA. ammarkheder.com/Otso
- cDESP, a B2B SaaS compliance platform for industrial refrigeration pressure equipment. Built by Wabel Group for DCRR. esp.dcrr.fr
- Pokemon Robot Auction (22 Dec 2023, Paris). The world's first Pokemon card auction co-hosted by a NAO humanoid robot. 1,693 viewers. Endorsed by the French Presidency.
- LingoRob (INRIA Bordeaux, 2021-2022). Crowdsourcing platform for multi-language linguistic corpora to train brain-inspired language models for human-robot interaction.

TEACHING

- BM40A1601 Foundations of AI and Machine Learning, Fall 2024 and Fall 2025 (LUT, blended, Lpr/Lahti).
- BM40A0202 Foundations of Computer Science, Spring 2026 (LUT, blended, Lpr).
- AMC-Lahti summer school, Application of AI / ML techniques in Atmospheric Science (11-15 August 2025, 25 participants).

ACADEMIC VISITS

- June 2026: research visit to the ARCHES team at Inria, hosted by Claire Monteleoni.
- May 2026 (5-6 May): two-day research stay with AMC member Prof. Jia Chen and her group at the Technical University of Munich (TUM). Start of a new international collaboration.
- July 2025: Istanbul Technical University. Air quality modelling with Dr Metin Baykara.
- April 2025: INRIA Bordeaux (N. Rougier and F. Alexandre) and Ecole Polytechnique Paris (V. Kalogeiton and M.-P. Cani).
- October 2025: LUMI training, Tallinn. Supercomputing, GPU profiling, performance optimisation.

TECHNICAL SKILLS

- Programming: Python (PyTorch, NumPy, scientific stack), C, SQL, JavaScript, HTML / CSS, VBA.
- HPC: distributed training on LUMI (up to 1,024 AMD MI250X GPUs), DDP, ROCm / CUDA, Slurm, Zarr.
- Deep learning: generative models (flow matching), vision transformers, recurrent and graph neural networks, physics-informed models; Lagrangian transport (FLEXPART-style) coupled to neural networks.
- Web and product: Django, Flask, full-stack web, B2B SaaS delivery.

LANGUAGES

- French: native.
- Arabic: native.
- English: fluent (working language at LUT and in research).